

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Analyzed the word-count distribution of tag, citation and evidence text, filtering statistical outliers with **IQR**
- Quantified duplication in the corpus, flagging **30%** of documents as identical with a de-duplication algorithm
- Served search over **600K+ evidence documents** from *Elasticsearch*, returning results in **under 1 second**
- Collaborated with **10 engineers** in an *Agile* team on a *Python (FastAPI)* platform drawing **10K+ site visitors**

Product Development Intern | Jetson

January 2025 – February 2025

- Tested hypotheses through **7 iterations of UI/UX testing** across **3 experimental features (100K+ users)**
- Translated testing results into a **5-page report** of strategies to optimize user engagement for Jetson's CEO
- Shortened the onboarding flow from **12 to 5 pages** in *Figma* and validated it via **A/B testing** with 30 test users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Ran exploratory analysis on **3 years** of TikTok data (**200+ posts**) with *Python (pandas)*, *Power BI* and *Excel*
- Discovered a content-mix skew: science “fun facts” posts drew **69.3%** of views from only **7%** of all content
- Recommended a “fun facts” content strategy to the Executive Director, who adopted it; engagement rose **24%**
- Completed an analytical project from problem definition through final presentation of findings to leadership

PROJECTS

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Ran a 4-person team on a pricing algorithm for **Kamion (YC S22)**; **2nd at UIUC Founder's Hackathon**
- Mined **15,000** scraped truck-listing images, auditing data quality with de-duplication and non-USD filtering
- Extracted condition, brand, model and year from photos with a *Claude Sonnet 4.6* agent and *JSON schema*
- Iterated on the schema after each wrong prediction until **10 consecutive images** came out correct on **3 features**
- Modeled truck prices with *quantile regression* on the extracted features, scoring **87% accuracy** on **30 samples**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Applied **4 Random Forest** models in a *Python FastAPI* back end to detect anomalies in water, waste and energy data from live readings and deliver actionable steps to fix those anomalies via a *Gemma* Voice LLM on a dashboard
- Created a **suite of 3 edge AI tools** that record a school's energy, water and food consumption for analysis
- Guided a team of **5** from ideation through development to presentation as a **USAII Global Hackathon Finalist**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Wrote *SQL* against a relational *Supabase (PostgreSQL)* database to slice papers by field, year and citation count
- Joined *OpenAlex* and *Semantic Scholar* data in an ingestion pipeline that resolves papers, then backfills metadata
- Visualized a paper's outbound citations across **26 research fields** in an interactive *React Flow* citation graph
- Shipped a **RAG** agent on *Claude Sonnet 4.5* that pulls from session-scoped vector documents in *Supermemory*

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Trained and validated a two-stage *EfficientNetV2-S* deep learning classifier (**4.39M** parameters) in *PyTorch* on **15K+ images**, sorting waste into 4 disposal pathways at **96.13% accuracy** and publishing a paper on *arXiv*
- Built a synthetic data generation pipeline (*U2-Net* segmentation, alpha-compositing) that generated **9K** images
- Released **2 original datasets** on *Kaggle* for the project and **placed 2nd at the Arizona State Science Fair**

SKILLS

Programming Languages: Python, SQL, Java, C++, JavaScript, HTML/CSS

Frameworks/Platforms: PostgreSQL, MySQL, Supabase, Elasticsearch, AWS, Google Cloud, FastAPI, Flask

AI/ML: scikit-learn, PyTorch, TensorFlow, Keras, Hugging Face, OpenCV, Claude API, Gemma 4, Grad-CAM

Data & Analytics: pandas, NumPy, A/B Testing, Power BI, Microsoft Excel, Hootsuite

Concepts/Practices: Hypothesis Testing, Behavioral Analysis, Engagement Analysis, Statistical Modeling